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AI & BIOMECHATRONICS
LABORATORY

# Development of an AI-Assisted Smart Wheelchair System Controlled by EOG Signals

**Mechatronic System Design Course Project**

Atalay Sunar , Çağlayan Emekli , İlayda Sarıkoca

**Project Advisor: Prof. Dr. Erhan AKDOĞAN**

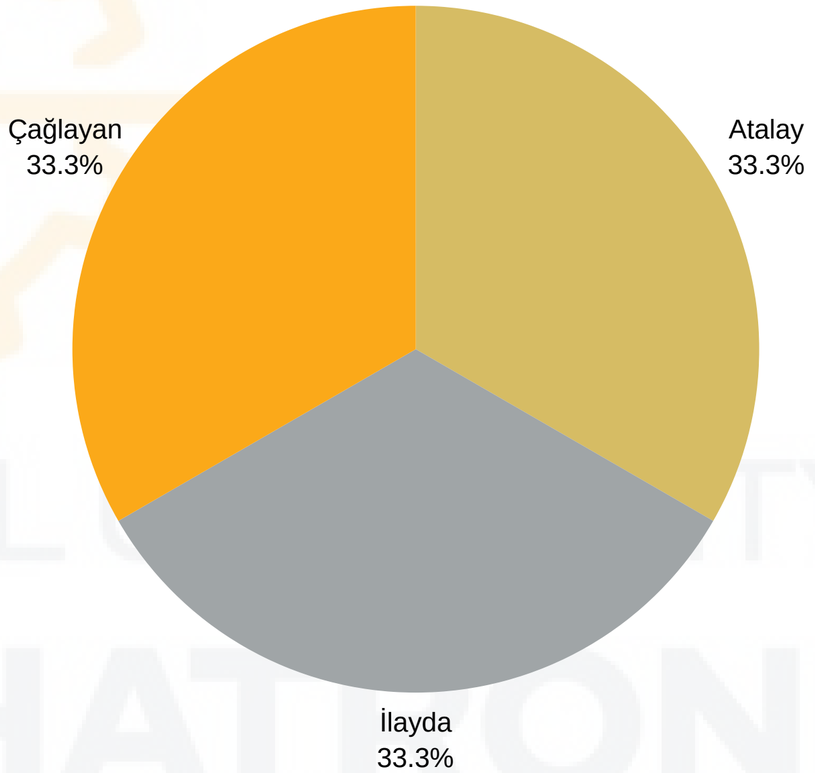
2025-2026

# Project Task Distribution

**Atalay:**
- Hardware Integration
- Motor Control & Power Systems
- Mechanical Assembly

**İlayda:**
- Software Development
- Signal Processing
- AI Model Development

**Çağlayan:**
- Data Collection & Analysis
- System Testing
- Electronics Design



# Problem

**Herhangi bir yardım almadan ya da yardımcı bir araç kullanmadan yürüyemeyenler, merdiven inip çıkamayanların cinsiyet ve yaş grubuna göre dağılımı, 2008-2022**

The percentage of individuals not able to walk, walk up and down stairs without any aids or assistances by sex and age group, 2008-2022

**[15+ yaş - age]**

(%)

| Yaş grubu / Age group | 2008 | | | 2010 | | | 2012 | | | 2014 | | | 2016 | | | 2019 | | | 2022 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Toplam Total | Erkek Male | Kadın Female | Toplam Total | Erkek Male | Kadın Female | Toplam Total | Erkek Male | Kadın Female | Toplam Total | Erkek Male | Kadın Female | Toplam Total | Erkek Male | Kadın Female | Toplam Total | Erkek Male | Kadın Female | Toplam Total | Erkek Male | Kadın Female |
| **Yürüyemeyenler** Not able to walk | 3,6 | 2,4 | 4,7 | 4,1 | 2,6 | 5,4 | 4,5 | 2,7 | 6,3 | 7,3 | 4,1 | 10,4 | 6,5 | 4,0 | 8,9 | 5,5 | 3,1 | 7,9 | 4,9 | 2,9 | 6,8 |
| 15-44 | 1,8 | 1,6 | 2,0 | 2,7 | 2,3 | 3,1 | 2,3 | 2,0 | 2,5 | 1,7 | 1,3 | 2,1 | 1,3 | 1,4 | 1,1 | 0,8 | 0,5 | 1,1 | 0,9 | 0,6 | 1,1 |
| 45-54 | 2,8 | 1,8 | 3,8 | 2,8 | 1,5 | 4,2 | 3,2 | 2,2 | 4,3 | 6,8 | 3,4 | 10,4 | 5,1 | 2,7 | 7,6 | 4,4 | 2,9 | 6,0 | 3,0 | 2,2 | 3,7 |
| 55-64 | 6,6 | 3,0 | 10,0 | 6,4 | 4,7 | 8,1 | 8,1 | 3,9 | 12,1 | 12,9 | 5,2 | 20,5 | 10,8 | 4,3 | 17,1 | 8,2 | 4,3 | 12,0 | 6,8 | 3,7 | 9,8 |
| 65-74 | 15,3 | 9,7 | 20,0 | 17,7 | 11,2 | 22,5 | 20,3 | 11,7 | 27,5 | 24,3 | 15,3 | 31,9 | 23,4 | 15,4 | 30,2 | 17,1 | 9,3 | 23,9 | 15,4 | 9,6 | 20,5 |
| 75+ | 34,0 | 31,7 | 35,6 | 37,7 | 25,0 | 48,2 | 39,9 | 29,9 | 46,8 | 51,2 | 38,3 | 59,7 | 48,0 | 36,0 | 55,8 | 44,7 | 32,0 | 53,1 | 37,3 | 25,8 | 45,0 |
| **Merdiven inip çıkamayanlar** Not able to walk up and down stairs | 3,8 | 2,4 | 5,2 | 4,8 | 3,1 | 6,5 | 5,3 | 3,1 | 7,4 | 9,0 | 5,1 | 12,8 | 8,7 | 5,0 | 12,4 | 7,9 | 4,6 | 11,1 | 6,6 | 3,7 | 9,4 |
| 15-44 | 1,6 | 1,4 | 1,9 | 2,9 | 2,5 | 3,3 | 2,8 | 2,2 | 3,5 | 2,4 | 1,7 | 3,2 | 1,9 | 1,7 | 2,2 | 1,5 | 1,0 | 2,0 | 1,4 | 1,0 | 1,9 |
| 45-54 | 2,7 | 1,8 | 3,6 | 3,7 | 1,7 | 5,6 | 3,5 | 2,1 | 4,9 | 9,0 | 3,9 | 14,2 | 8,8 | 4,2 | 13,4 | 7,5 | 4,5 | 10,6 | 5,1 | 2,8 | 7,3 |
| 55-64 | 7,5 | 3,6 | 11,2 | 8,6 | 5,1 | 11,9 | 10,8 | 5,2 | 16,1 | 16,3 | 7,1 | 25,2 | 15,4 | 6,5 | 24,1 | 12,9 | 6,7 | 18,9 | 10,3 | 4,9 | 15,5 |
| 65-74 | 17,5 | 9,8 | 23,8 | 21,2 | 12,8 | 27,4 | 22,8 | 13,2 | 30,6 | 30,7 | 19,6 | 40,1 | 29,3 | 18,6 | 38,5 | 24,7 | 13,6 | 34,3 | 18,9 | 11,3 | 25,6 |
| 75+ | 35,8 | 29,2 | 40,3 | 42,8 | 33,2 | 50,7 | 44,5 | 34,7 | 51,2 | 53,6 | 42,1 | 61,3 | 55,4 | 39,8 | 65,7 | 50,8 | 39,0 | 58,7 | 43,4 | 30,9 | 51,6 |

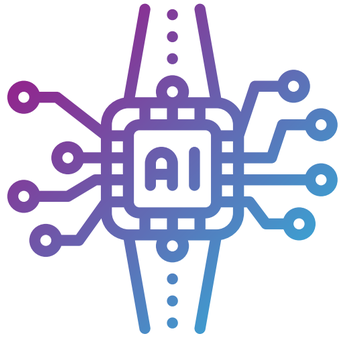
**TÜİK, Türkiye Sağlık Araştırması, 2008, 2010, 2012, 2014, 2016, 2019, 2022**

TurkStat, Türkiye Health Survey, 2008, 2010, 2012, 2014, 2016, 2019, 2022

## Current Situation

- Traditional wheelchairs require hand or arm movement.
- Paralyzed individuals cannot use them independently.
- Eye-tracking systems are expensive and inefficient in low-light conditions.
- EOG-based accessible control systems are limited.

# Solution




EOG signals are processed using artificial intelligence and converted into wheelchair movement commands.



The system provides full independence to individuals with upper body movement limitations without the need to use their hands.



User experience and safety are enhanced through a mobile interface and an emergency stop feature.

# User Needs to System Design Mapping

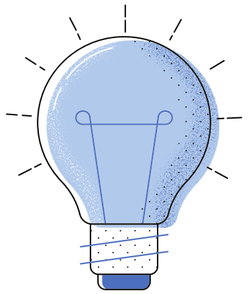
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| 1. Project Goal | 2. User Needs (Qualitative) | 3. Technical Requirements | 4. System Design | 5. Justification & Next Steps |
|---|---|---|---|---|
| Enable hands-free wheelchair control using eye movements. | User cannot use hands; control must be simple and safe. | Classification Accuracy ≥95.9%, latency <1s, false command <5%. | EOG Sensor & Electrodes → ESP32-S3 → PC-Based Signal Processing | EOG is non-invasive and robust. Next: classification tests. |
| Safe execution of basic motion commands. | Unintentional movement must be avoided. | Dedicated STOP command, high activation threshold. | Look Left/Right → Turn, Double Blink → Forward, Long Blink →STOP/forward | Command mapping minimizes false activation. |

# User Needs to System Design Mapping

| | | | | |
|---|---|---|---|---|
| Sufficient operating time. | Battery should last long enough for daily use. | 12 V system, nominal current <40 A, ≥60 min nominal, ≥30 min worst-case. | 12 V 24 Ah (or equivalent) gel battery, BTS7960, MY1016Z motors. | Higher Ah improves stability and battery life. Next: endurance test. |
| Smooth and safe motion. | No sudden acceleration. | Max speed ≤1.5 m/s, limited angular velocity. | PWM motor control, speed limiting, ramp control. | Safety-oriented motion profile. |
| Low cost and reproducibility. | Components should be affordable and available. | Commercial off-the-shelf components. | Arduino, standard EOG board, ready motor drivers. | Cost-effective and easy to replicate. Next: PCB integration. |

# Project Objective and Technical Goals



**Project Objective:**
To enhance the independence of people with disabilities through an AI-assisted system using advanced sensor technologies.





**Technical Goals:**

- Develop a reliable EOG signal acquisition system.
- Implement AI-based control logic.
- Ensure low-cost and energy-efficient operation.

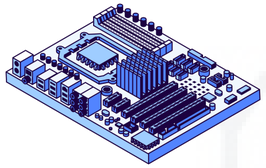
# METHOD



**Signal Acquisition & Processing**
- EOG electrodes → raw data collection
- Noise filtering & feature extraction



**Model Development**
- EOG dataset creation
- ANN model



**Hardware Design**
- ESP32/Arduino integration
- Motor driver, power circuit, sensors



**Software Development**
- Real-time classification
- Control algorithm



**System Integration & Testing**
- Full wheelchair integration
- Performance tests



# Technical Specifications & System Requirements

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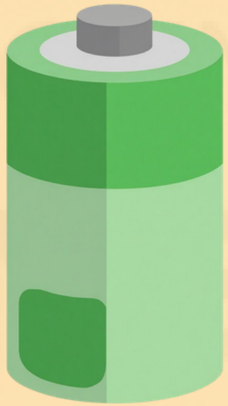
### Performance

- Command Response Time: < 1s
- Classification Accuracy: > 95.9%



### Power

- Battery Life: > 1 hour
- Operating Voltage: 12V DC



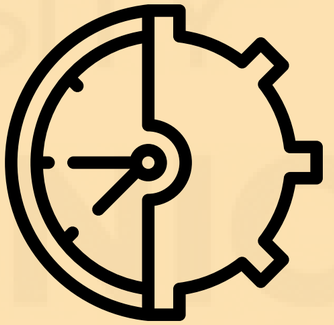
### Safety

- Obstacle Detection: < 0,4 m
- Emergency Stop: Button + Auto Stop
- Max Speed: 0.5-0.8 m/s (1.8-2.9 km/h)

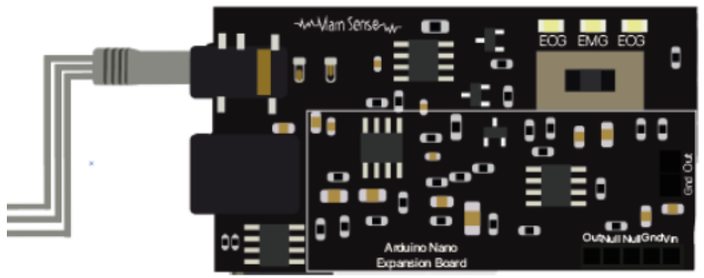


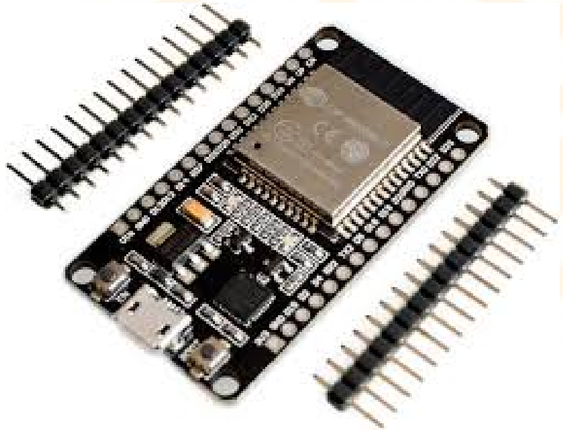
### Usability

- Calibration Time: < 10 seconds
- Number of Commands: 5 (Left, Right, Forward, Stop)



# General System Architecture





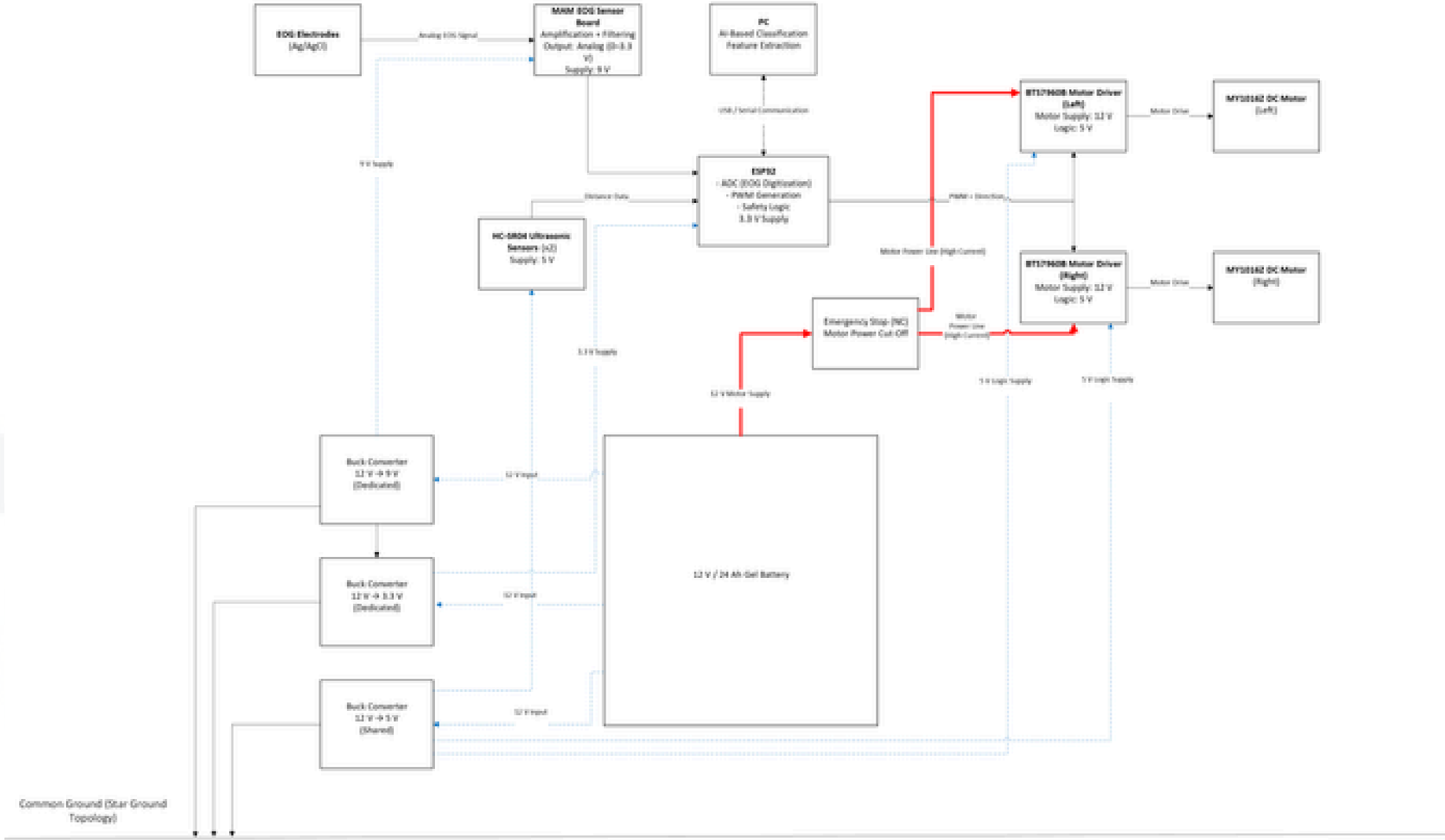
**Acquisition**: EOG Signals via Electrodes.

**Processing**: Signal Filtering & AI Classification.

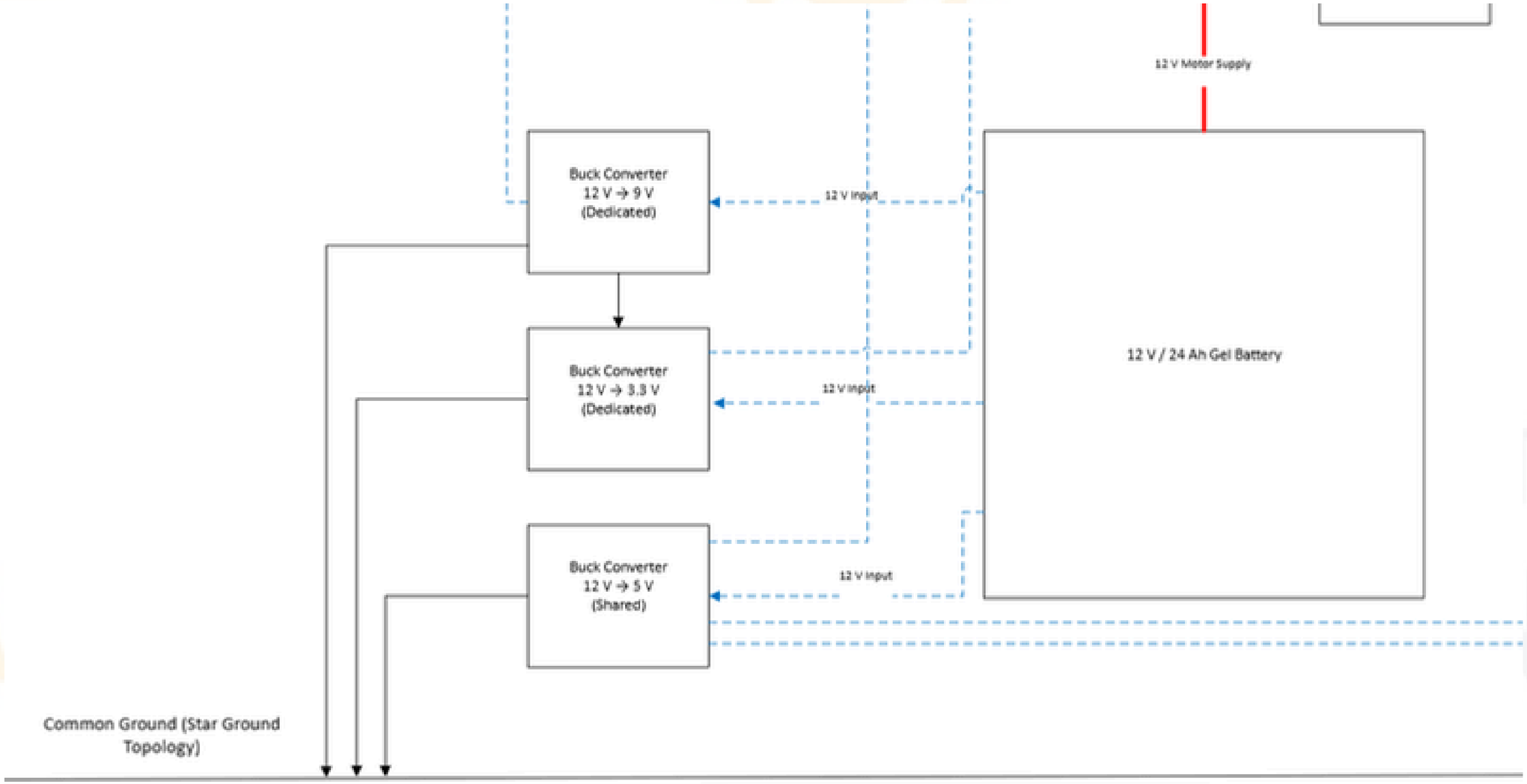
**Control:** Microcontroller Logic.

**Actuation:** Dual MY1016Z Motor Drive.

# General System Architecture

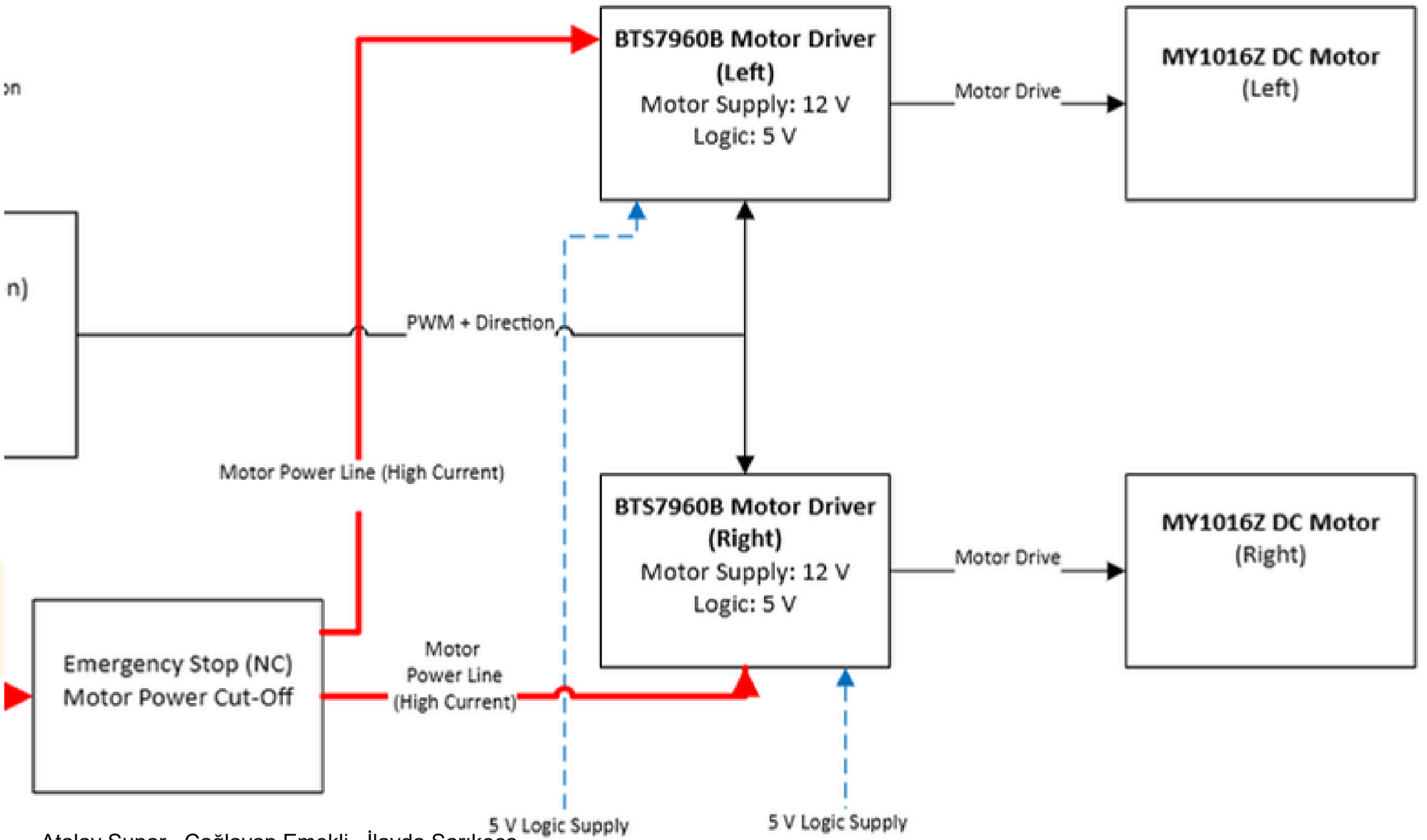


# Power Management and Distribution

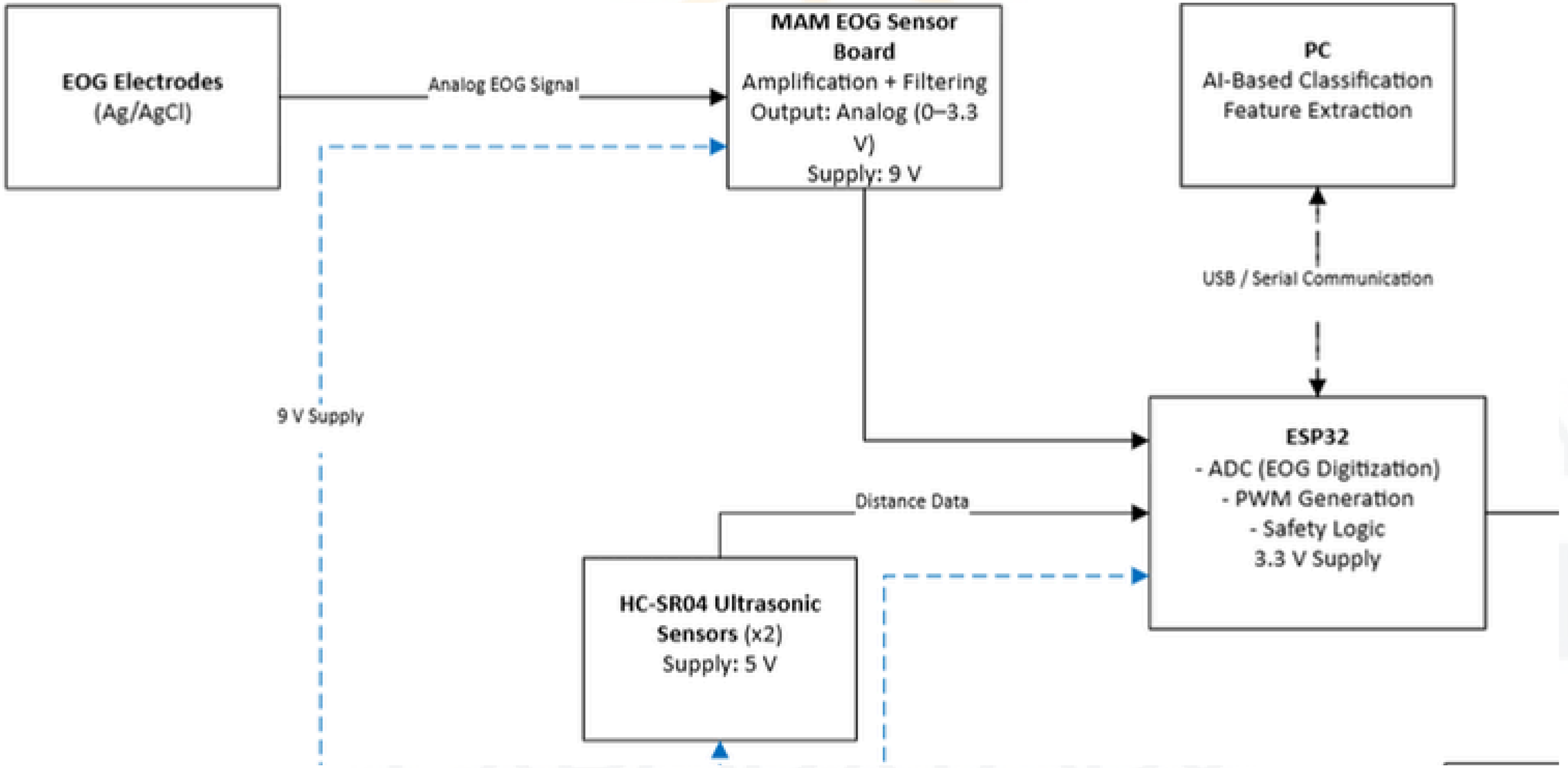



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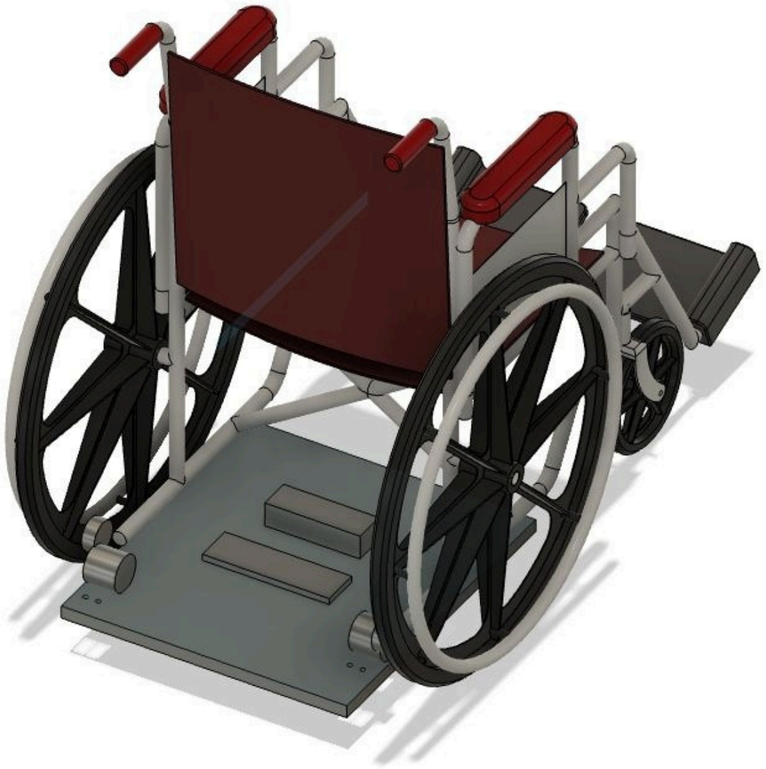
# Actuation &Emergency Stop



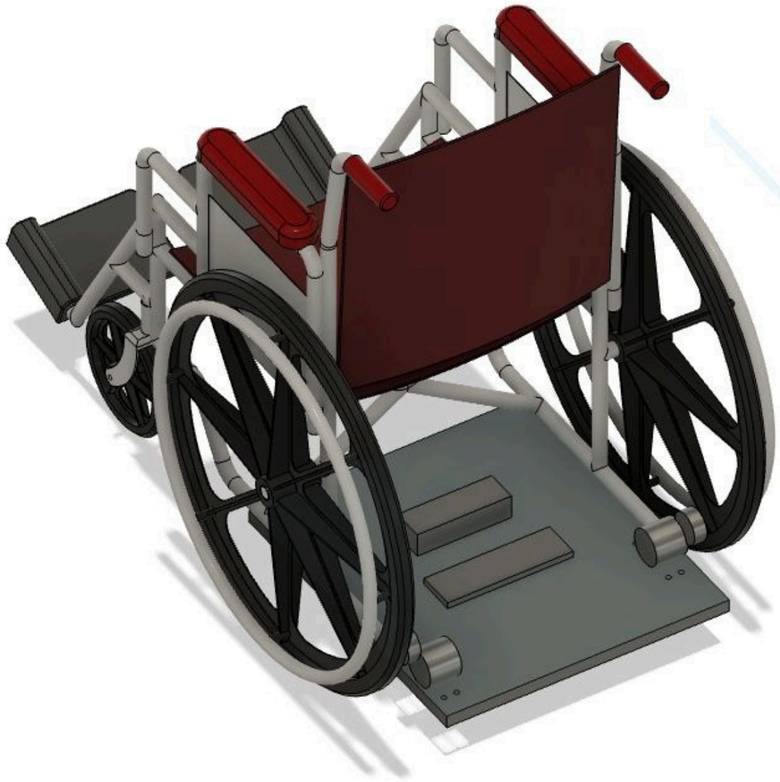
# Signal Acquisition Interface and Artificial Intelligence Architecture



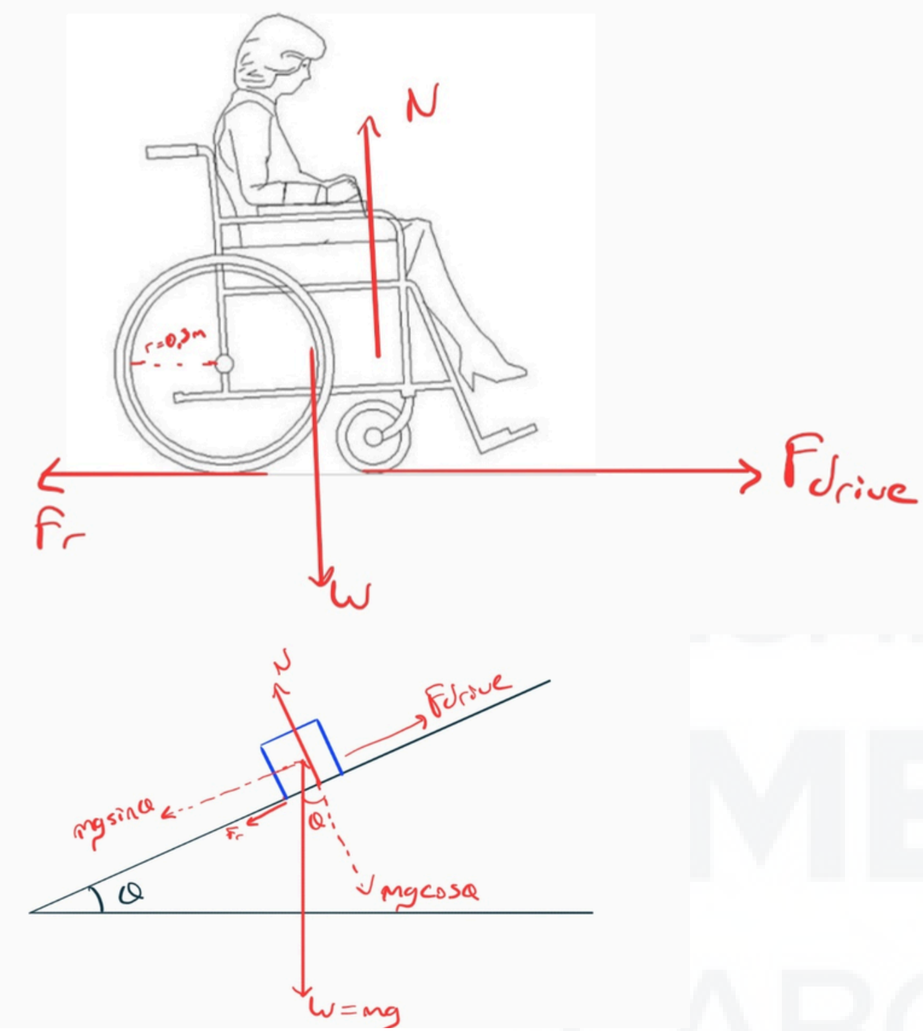
# Mechanical design & platform integration







# Mechanical Load Analysis




| Parameter | Symbol | Value |
|---|---|---|
| Total Gross Mass | m | 110 kg |
| Wheel Radius | r | 0.30 m |
| Incline Angle | θ | 5∘ |
| Rolling Resistance | Fr | 21.58 N |
| Total Drag Force (5∘) | Ftotal | 115.6 N |
| Required Torque | τtotal | 34.7 Nm |

# Actuator Selection & Performance Verification

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Design Scenario: Climbing a 5° incline with a total mass of 110 kg.

**Force Calculation:**

$$F_{slope} = m \cdot g \cdot \sin(5^\circ) \approx 94.0\,N.$$

$$F_{rolling} = \mu_r \cdot m \cdot g \cdot \cos(5^\circ) \approx 21.5\,N.$$

$$TotalResistance(F_t) :\approx \mathbf{115.5\,N}.$$

**Torque Requirement:**

$$\tau_{total} = F_t \cdot r = 115.5\,N \cdot 0.3\,m = 34.7\,Nm.$$

Required Torque per Motor:: $\mathbf{17.4\,Nm}$.



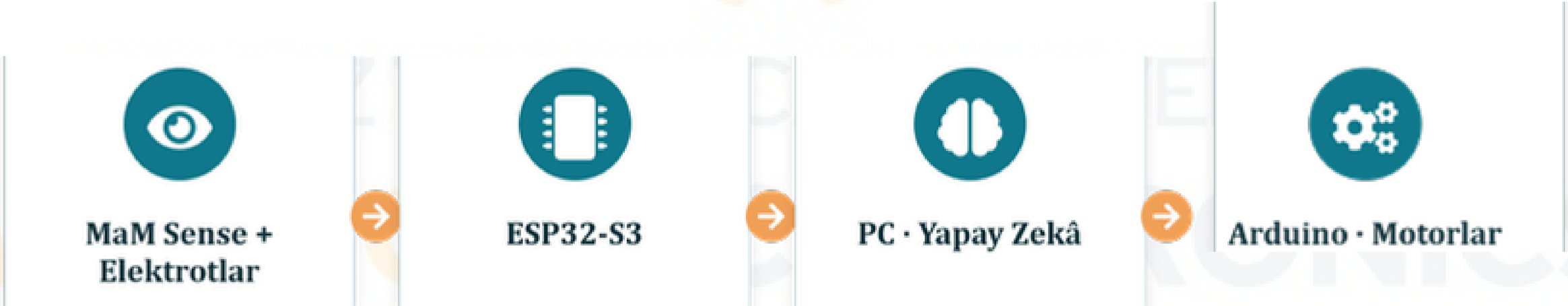
**Type:** Brushed DC Geared Motor.
**Rated Power:** 250W at 12V.
**Gear Ratio:** 9.7:1 Internal Reduction.

We chose this motor for its high torque, safe speed control, and system compatibility.

# AI BLOCK DIAGRAM – EOG-Controlled Smart Wheelchair

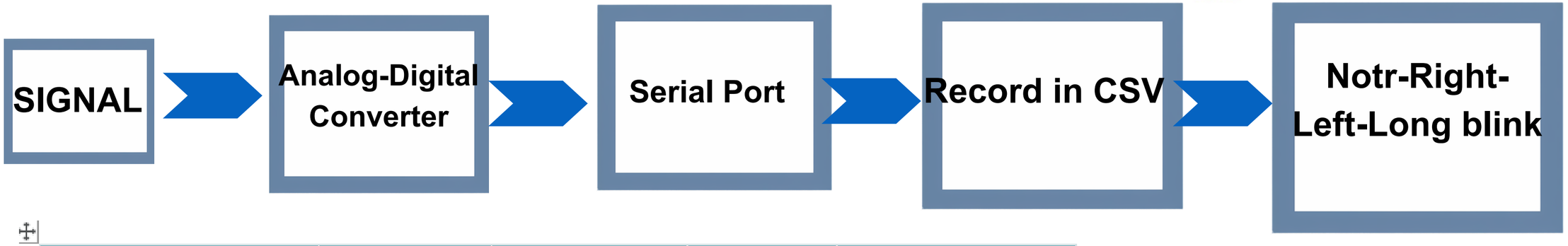
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# Collecting Data

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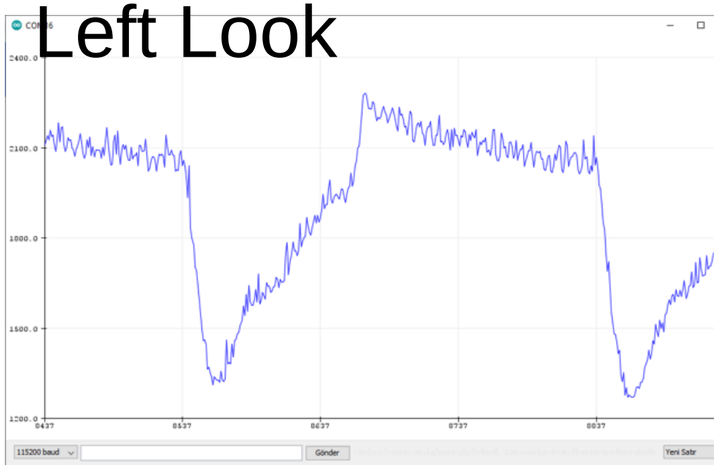



| Class | Events | Windows | Duration | Raw samples |
|---|---|---|---|---|
| Neutral / Nötr | 207 | 207 | 60 s | ~21,001 |
| Right / Sağ | 20 | 99 | 60 s | ~21,002 |
| Left / Sol | 33 | 165 | 60 s | ~21,002 |
| Long-Blink / U. Kırpma | 36 | 176 | 60 s | ~21,002 |

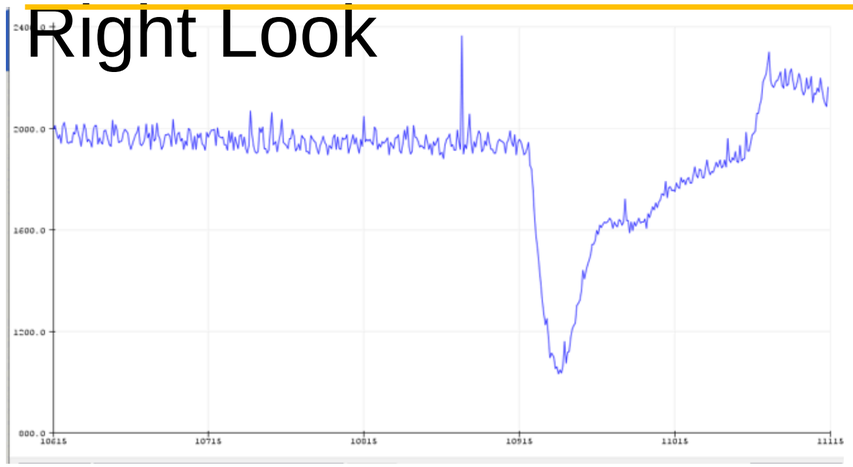
45

$$f_s = \frac{21{,}001\text{sample}}{60sn} = 350 \text{ samples per second} = 350Hz$$

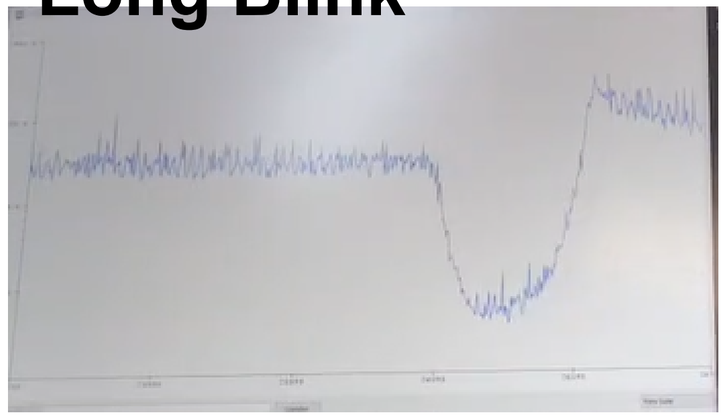
## Left Look



## Right Look

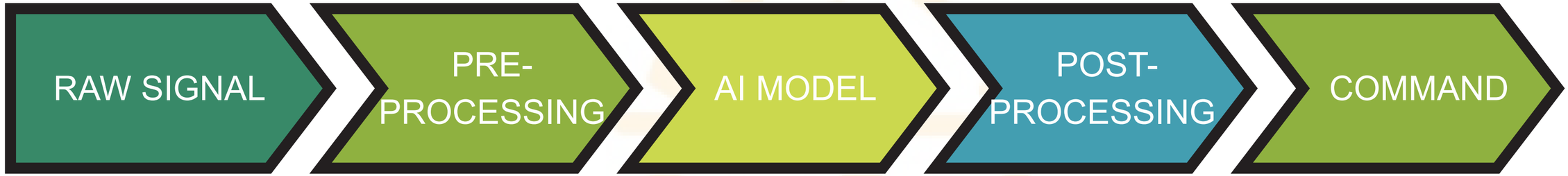


## Long Blink



# Pre-processing and post-processing






**Why is pre-processing needed?**

- Signals can have different lengths
- Each signal sits at a different “height”.
- One person's signal may be strong, another's weak.

**Why is post-processing needed?**

- Raw AI predictions can be uncertain or incorrect, posing direct safety risks in physical wheelchair control.
- Post-processing acts as a safety grid, using confidence thresholds and voting to filter out these errors.
- It ensures reliability by converting probabilistic model outputs into safe, deterministic motor commands.

# Pre-processing

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## PRE-PROCESSING

**0 Catch the event (in real time)**

The system continuously monitors the stream. A ~1.1-second window is clipped only when a sudden peak exceeds a set threshold (THR_OLAY = 350).

**1 Resample — bring every signal to 200 points**

Regardless of the hardware sampling rate, every clipped segment is standardized to exactly 200 points (L=200).

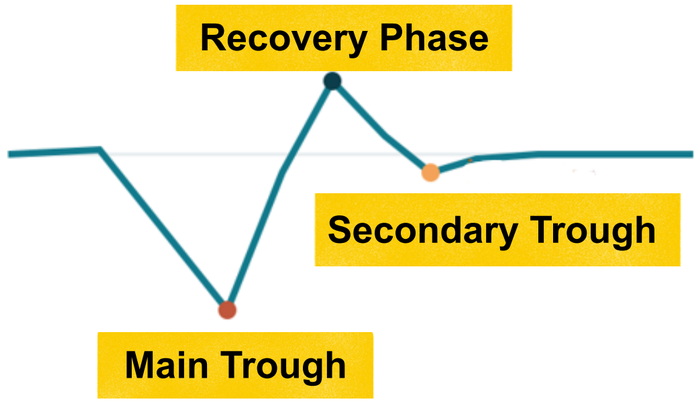
**2 Remove the baseline**

The mean is subtracted (s=s-mean(s)) to center the wave around 0. Absolute DC offsets are discarded to focus purely on the waveform shape.

**3 Z-Score Normalization**

The signal is divided by its standard deviation (s = s/std(s)). This scales amplitudes uniformly, neutralizing variation between weak and strong signals.

### Raw EOG Window

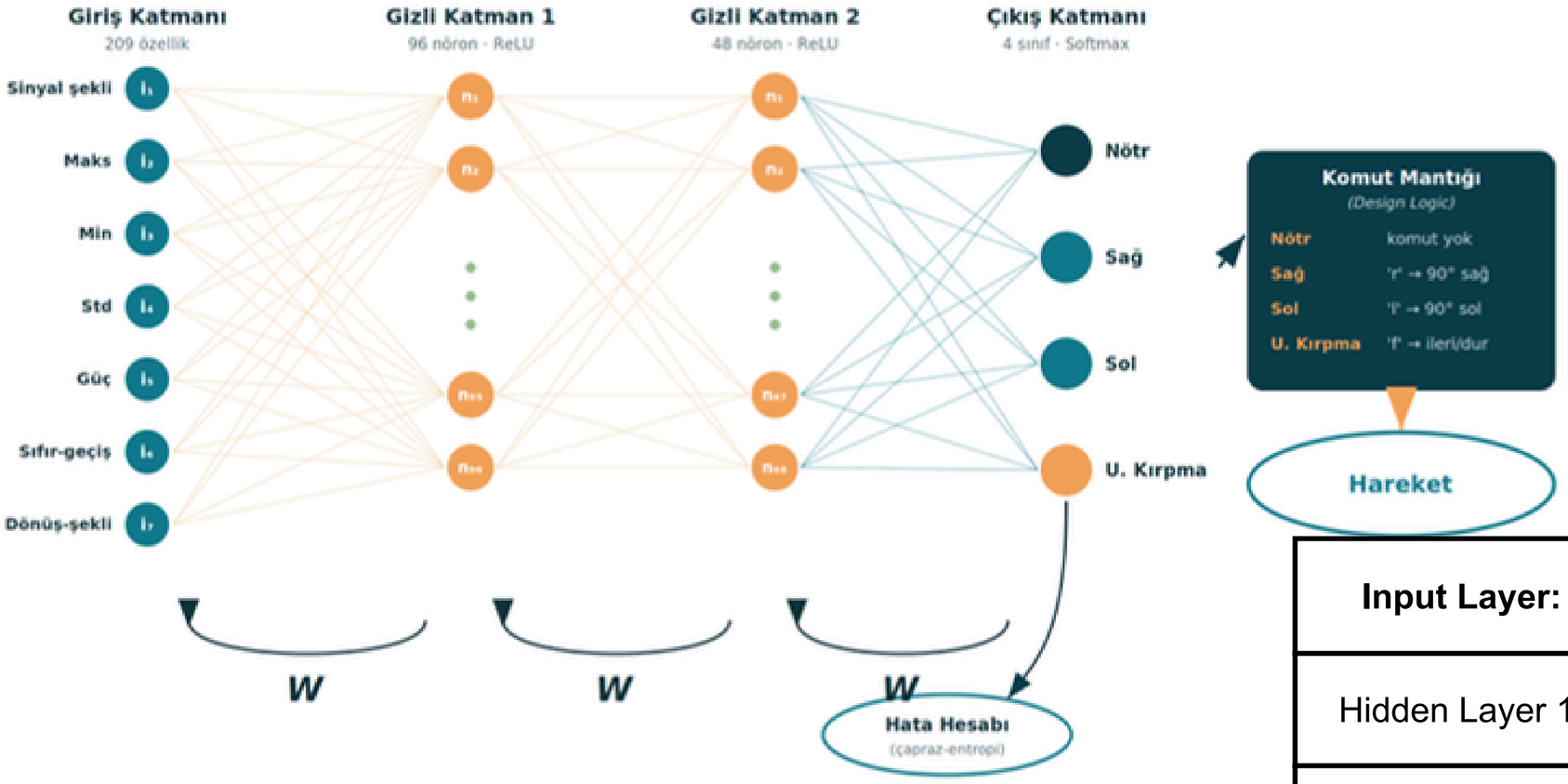



| Feature Group | Dimension | Description / Function |
|---|---|---|
| **Signal Waveform** | 200 Values | The normalized amplitude values of the resampled signal. |
| **Statistical Summary** | 5 Values | **1.Peak 2.trough 3.standard deviation 4.energy (power) 5.zero-crossing rate.** |
| **Return-Shape** | 4 Values | **The key to Right/Left separation:** Analyzes the "tail" or recovery phase after the main trough (secondary trough depth, prominence, maximum derivative, and roughness). |
| **TOTAL** | **209** | **The complete input vector fed into the network.** |

# Creating the Multilayer Perceptron (MLP)

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Our model is designed as a fully-connected feed-forward Multilayer Perceptron (MLP), structured to optimize feature extraction and classification.




| | |
|---|---|
| **Input Layer:** | 209 Nodes (corresponding to the 209 input features) |
| Hidden Layer 1: | 96 Neurons |
| Hidden Layer 2: | 48 Neurons |
| Output Layer: | 4 Neurons (corresponding to the 4 target classes) |

# Configure the Network

| Hyperparameter | Value | Role / why |
|---|---|---|
| **Hidden layers** | 96, 48 (ReLU) | Model capacity; narrowing funnel |
| **Output** | Softmax (4) | Class probabilities |
| **Optimiser** | Adam | Adaptive, stable, little tuning |
| **Learning rate** | 0.001 | Step size (Adam init) |
| **L2 reg. (alpha)** | $1\times10^{-4}$ | Overfitting control |
| **Batch size** | auto (≤200) | Mini-batch updates |
| **Loss** | Cross-entropy | Multiclass objective |
| **Max iterations** | 1800 | Upper limit |
| **Convergence** | tol $1\times10^{-4}$ · n_iter_no_change 10 | Stopping rule |

# Post-processing & Safety Filters

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## Post-processing

1. **Probability Selection (Argmax)** — The system identifies the class with the highest probability score.
2. **Confidence Threshold** — If the top prediction score falls below 55%, the output is discarded. No command is executed during ambiguous states (Safety First).
3. **Temporal Smoothing** — The system retains a rolling window of the last 3 predictions and applies a majority vote to eliminate sporadic outliers.
4. **Command Logic** — Once a command is issued, a new command cannot be sent until the system registers a NEUTRAL state. This state machine prevents the wheelchair from locking into an infinite turn.
5. **Final Motor Execution Mapping**

**Final Motor Execution Mapping**

| | |
|---|---|
| Notr | Sends 'n' code |
| Right | Sends 'r' code |
| Left | Sends 'l' code |
| Long Blink | Sends 'f' code |

# AI Classification Performance

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| Actual ↓ / Pred → | Neutral | Left | Right | Long-Blink |
|---|---|---|---|---|
| **Neutral** | 61 | 0 | 0 | 0 |
| **Left** | 0 | 39 | 0 | 1 |
| **Right** | 0 | 1 | 23 | 1 |
| **Long-Blink** | 0 | 5 | 0 | 66 |

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| **Neutral** | 1.000 | 1.000 | 1.000 | 61 |
| **Left** | 0.867 | 0.975 | 0.918 | 40 |
| **Right** | 1.000 | 0.920 | 0.958 | 25 |
| **Long-Blink** | 0.971 | 0.930 | 0.950 | 71 |
| **Macro avg** | 0.959 | 0.956 | 0.956 | 197 |

## Gecikme Analizi

Göz hareketinden motor tepkisine — bileşen bazında

| | |
|---|---|
| ESP32 → PC (seri) | < 1 ms |
| Jest yakalama + ortalama | ~0.55 sn |
| Oylama kararlılığı | ~0.45 sn |
| Çıkarım (model) | ~1 ms |
| PC → Arduino (komut) | ~2 ms |

Toplam (göz → komut)

**~0.6–1.0 sn**

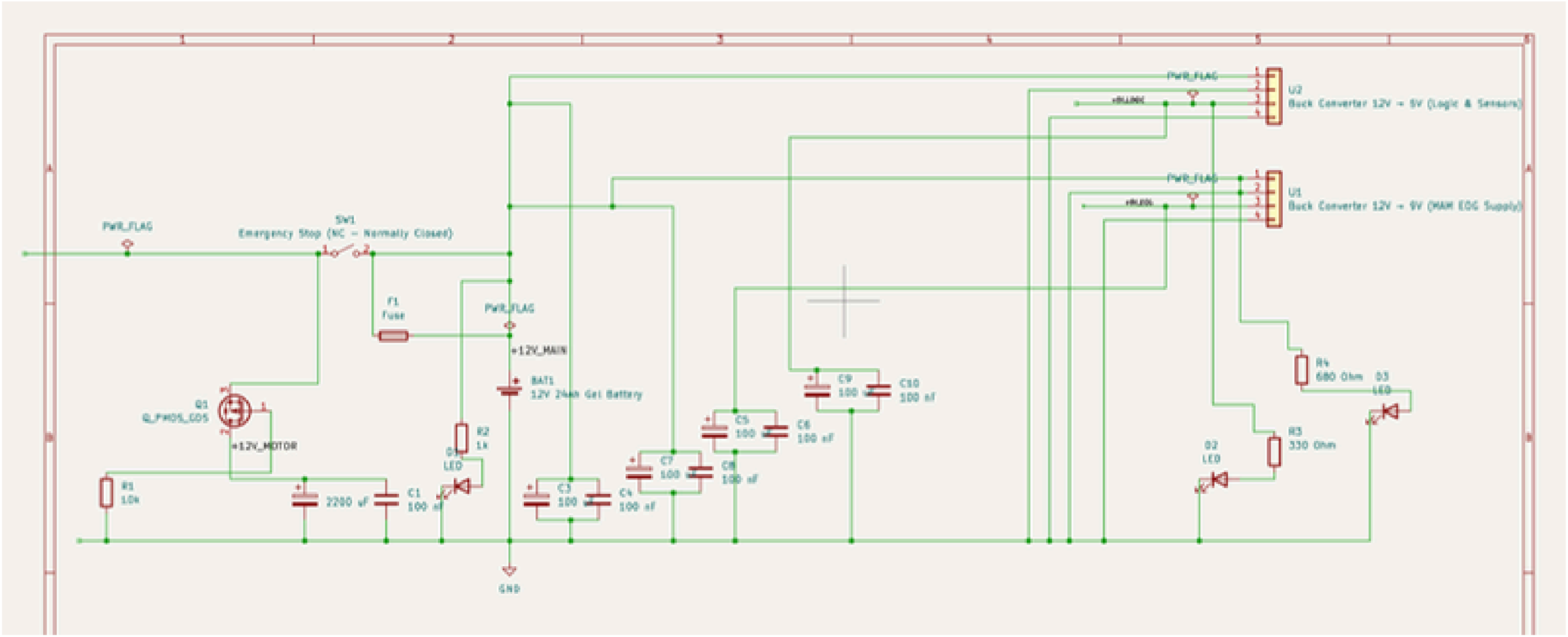
*Çoğunluğu yakalama+oylama*

**Accuracy Rate**

**%95.9**

# Power Management System & Voltage Regulation

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AI & BIOMECHATRONICS LABORATORY




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# Power Management & Energy Source






## 12V 24Ah Gel Battery

**1. Current Demand Calculation**

**Motor Power:** 2 × 250W = **500W**

**Maximum Current:** $Imax = 500W \div 12V \approx 41.6A$

**Design Current:**

Set at 45A to include safety margin.

**2. Capacity Selection**

- Average Consumption:
- ≈ 22.5A during typical usage (50% of design current).

Target Capacity:

- 22.5Ah for 1 hour of operation.

Selected Battery:

- 24Ah gel battery for sufficient buffer.

# Control Hub & Signal Acquisition

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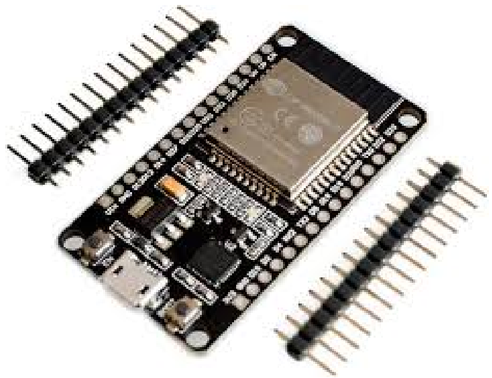
**Processor:** ESP32-S3 (Dual-core 240 MHz, 12-bit ADC).
**System Role:** Executes real-time ANN Classification and Motor Control logic.
**Data Integrity:** 12-bit resolution for high-fidelity EOG signal sampling.
**Noise Mitigation:** Star Ground Topology and regulated power rails to ensure signal quality

### Dönüş — MPU6050 jiroskop

Encoder tık sayımı tekerlek kaymasında kümülatif hata biriktirir; 90° set-noktasına güvenilir yakınsama sağlanamadı.

Çözüm: MPU6050 I2C IMU'dan yaw açısı doğrudan okunur (G(s)=0.00909/(s+0.045) model). Ölçülen yaw ≥ 90° koşulu sağlanınca motor durdurulur — yüzey bağımsız tekrarlanabilir dönüş.

3 sn güvenlik zaman-aşımı (watchdog) + dönüş sırasında acil durdurma komutu öncelikli dinlenir. NC buton: motor sürücü besleme hattında seri (IEC 60204-1 Kategori-0 durma).



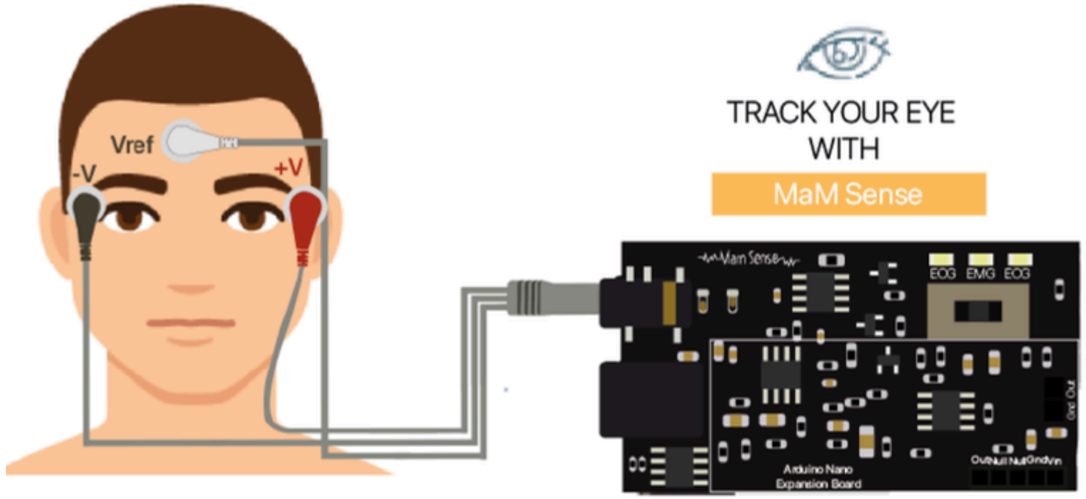
### Düz gitme — bağımsız hız kontrolü

BTS7960 H-köprüsü (43 A tepe, 20–25 kHz PWM) her tekere bağımsız sürücü. Encoder periyot-bazlı hız ölçümü: $v \propto 1/T_tick$. PID (Kp=6, Ki=3, Kd=1); her teker bağımsız 0.30 m/s set-noktasına kilitlenir.

Diferansiyel sürüş: $v_R = v + (L/2)\cdot\omega$, $v_L = v - (L/2)\cdot\omega$. Dönüşte MPU6050 yaw geribildirimli kapalı çevrim.

İki teker aynı hızda → doğrusal yörünge. Asimetrik sürtünme/motor farkı kapalı çevrimle otomatik telafi edilir.

# EOG Interface: MaM Sense Board




- **Front-End:**High-performance biopotential acquisition board.
- **Hardware Amplification:** Total Gain ≈ 1260x (Brings µV signals to 0-3.3V range)

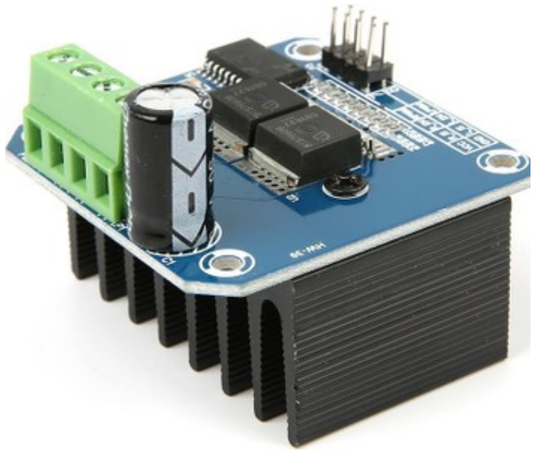
**Hardware Filtering:**

- **HPF (0.1 Hz):** Removes DC offset and baseline drift.
- **LPF (6.2 Hz):** Eliminates EMG (muscle activity) and environmental noise.
- **Output:** Stable analog EOG signal for ESP32 digitization.

# Actuation & Safety Mechanisms







- **Motor Drivers:** Dual BTS7960 H-Bridges (Supports up to 43A peak current).
- **Propulsion:** Two MY1016Z DC Motors (High torque/low RPM for 110 kg payload) **.**

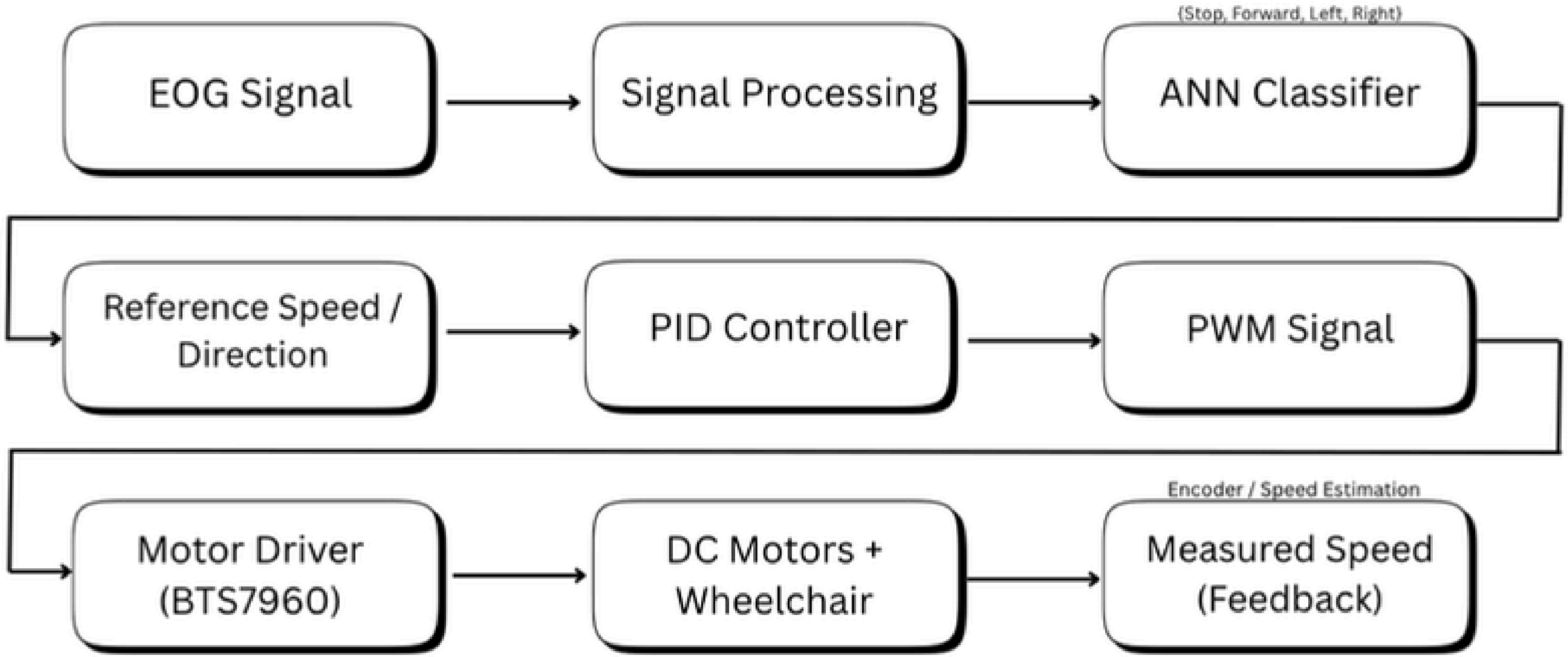
**Safety Interlocks:**

- **Hardware:** Physical Emergency Stop to cut motor power immediately.



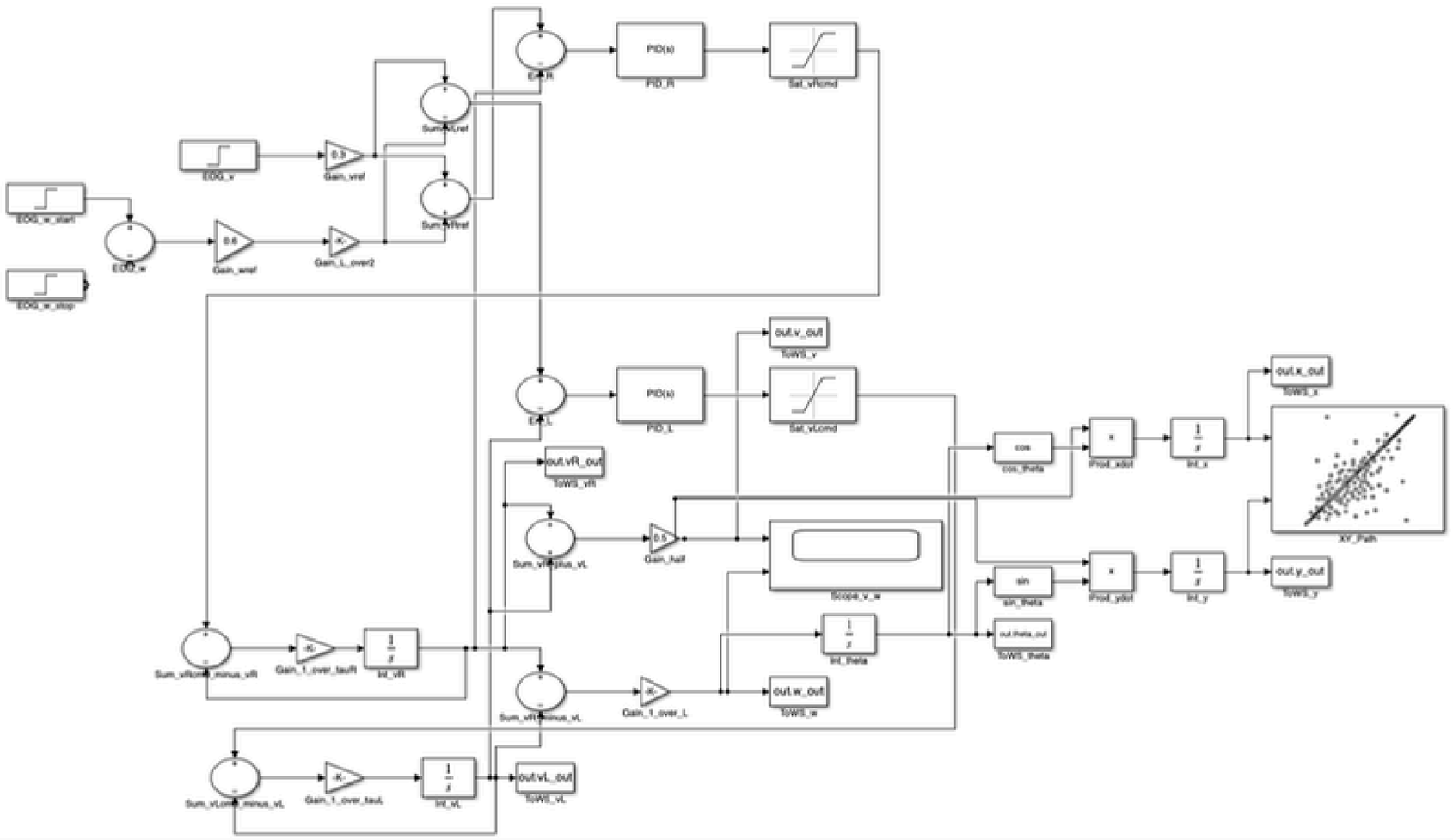
# Control Architecture of the EOG-Controlled Wheelchair





# Simulink Model of the EOG-Controlled Differential Drive Wheelchair

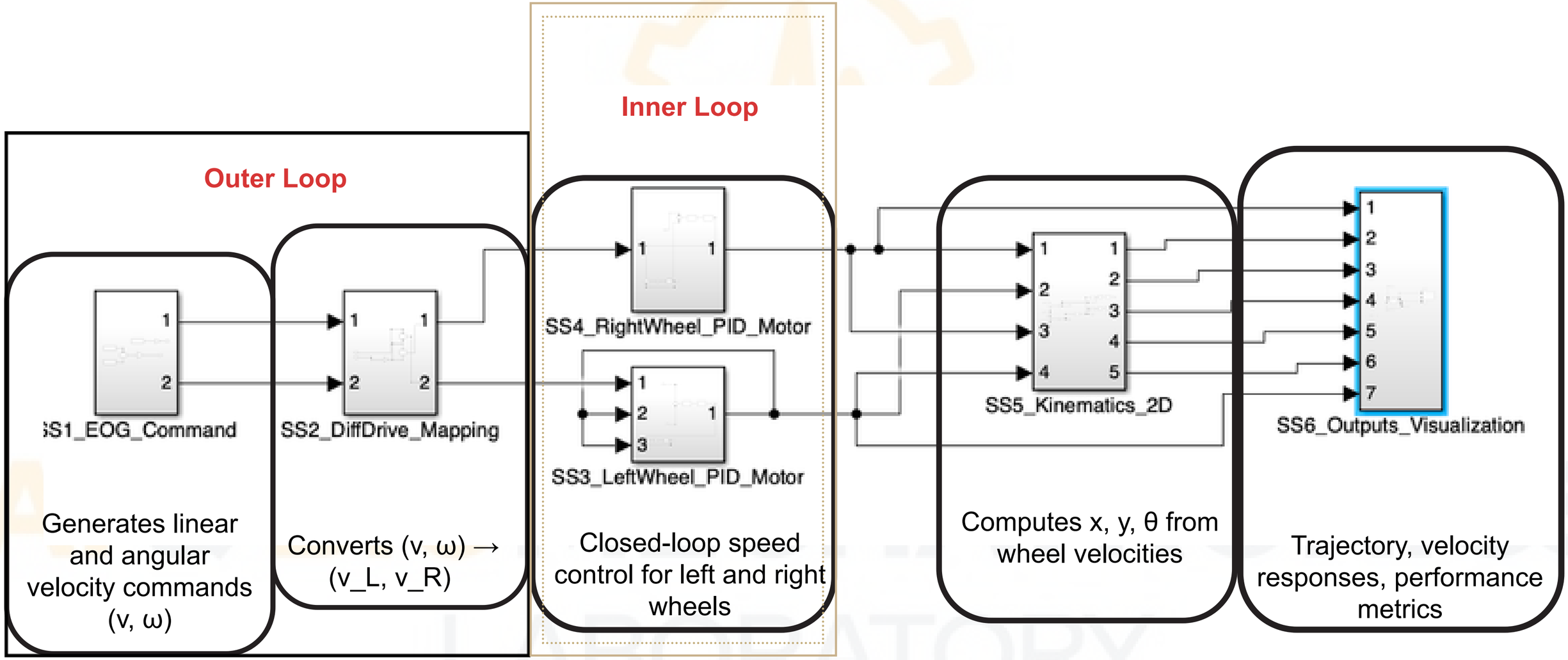
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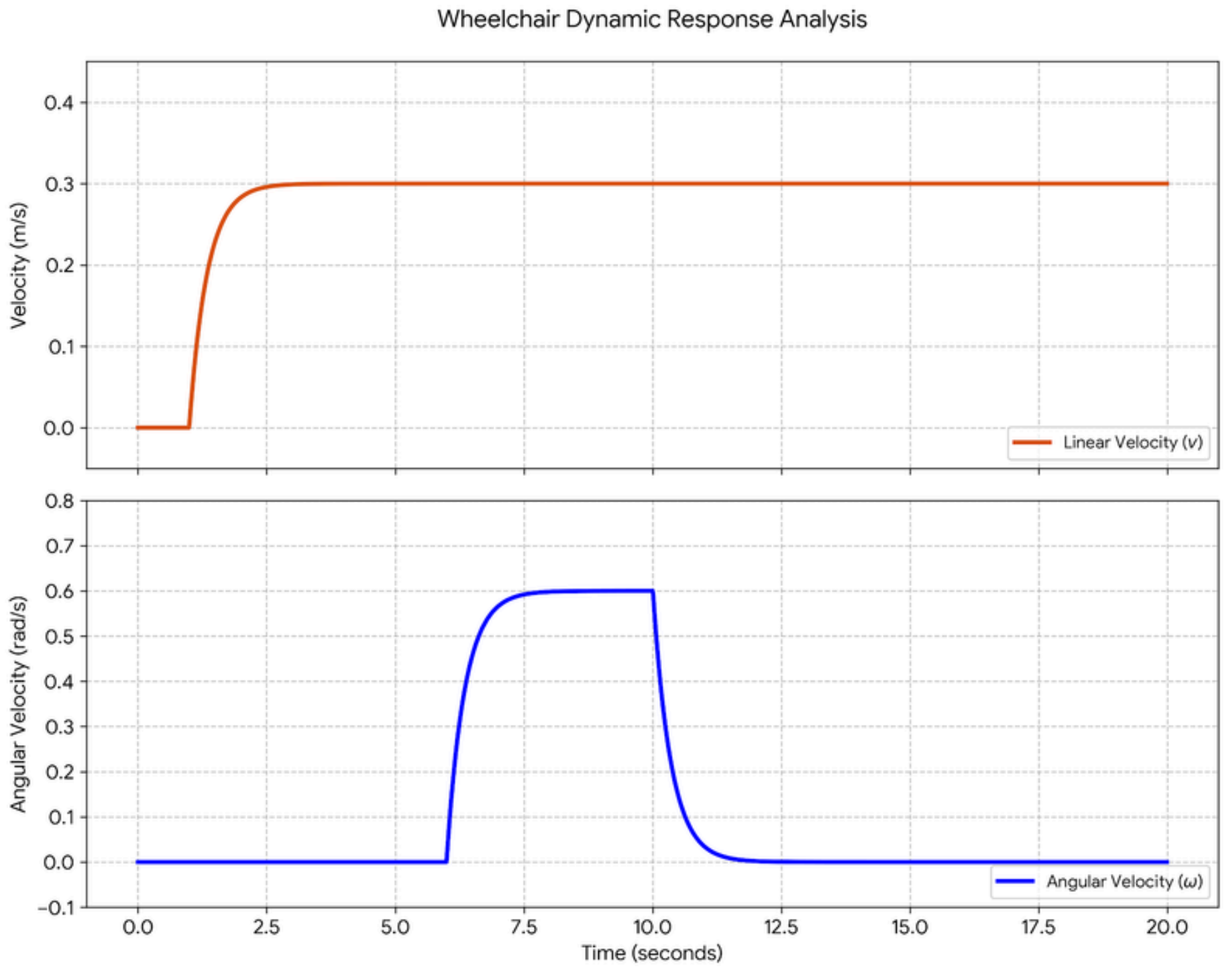

## Simulink Model Description

- **EOG-Based Command Generation**
  Linear (v) and angular (ω) velocity references are generated from EOG signals.
- **Differential Drive Mapping**
  Velocity references are converted into left and right wheel speed commands.
- **Closed-Loop Wheel Speed Control**
  Each wheel is regulated independently using a PID controller.
- **Kinematic Modeling**
  Wheel velocities are used to compute planar motion (x, y,θ).

# Overall Cascaded Control Structure (Simulink Model)



# SIMULATION RESULTS AND PERFORMANCE ANALYSIS




## Velocity Response (Control Performance)

**Wheelchair Velocity Response under EOG Commands**

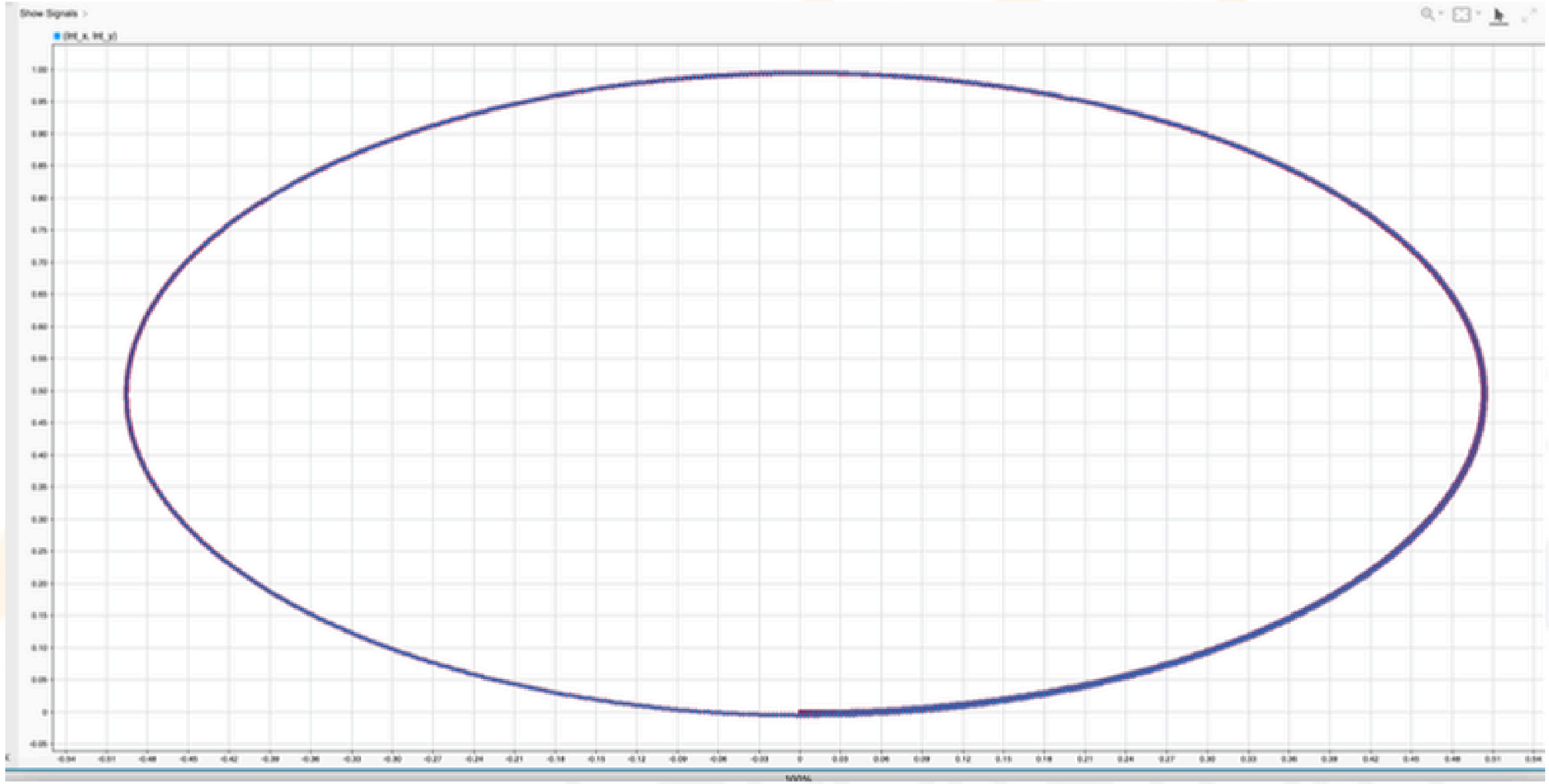
- Linear velocity (v) is limited to 0.3 m/s
- Angular velocity (ω) is activated between t = 6–10 s
- Smooth acceleration achieved via PID control
- No overshoot observed (user comfort & safety)

PID Controller Parameters:
Kp = 6
Ki = 3
Kd = 1

# SIMULATION RESULTS AND PERFORMANCE ANALYSIS

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## 2D Trajectory (Kinematic Validation)




**2D Trajectory of the Differential Drive Wheelchair**

- Trajectory generated using differential-drive kinematics
- Path curvature caused by angular velocity command
- Continuous and smooth motion without instability

$$\dot{x} = v\cos(\theta)$$
$$\dot{y} = v\sin(\theta)$$
$$\dot{\theta} = \omega$$

# SIMULATION RESULTS AND PERFORMANCE ANALYSIS

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## System-Level Interpretation



**System-Level Motion Behavior**

- Initial straight motion (v command only)
- Curved motion during turning phase
- Return to straight trajectory after $\omega \rightarrow 0$

# System Parameters

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| **Total Mass** | M | 110 | kg |
| **Wheel Radius** | r | 0.3048 | m |
| **Chassis Width** | L | 0.6 | m |
| **Motor Rated Torque** | $\tau_{nom}$ | 1 | Nm |
| **Motor Rated Current** | $I_{nom}$ | 20 | A |
| **Torque Constant** | $K_t$ | 0.05 | Nm/A |

# Kinematic Modeling

The wheelchair operates under Differential Drive Kinematics. The global pose is defined as vector $q = [x, y, \theta]^T$

## Linear and Angular Velocity Equations:

$$v = \frac{r(\omega_R + \omega_L)}{2} \quad , \quad \dot{\theta} = \frac{r(\omega_R - \omega_L)}{L}$$

## State-Space Representation:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}$$

# Coupled Dynamics (110 kg Payload)

Modeling the interaction between the two motors and the total system mass.

## 1. Equation of Motion:

$$\begin{bmatrix} \tau_L \\ \tau_R \end{bmatrix} = \begin{bmatrix} A & B \\ B & A \end{bmatrix} \begin{bmatrix} \dot{\omega}_L \\ \dot{\omega}_R \end{bmatrix} + \begin{bmatrix} K & 0 \\ 0 & K \end{bmatrix} \begin{bmatrix} \omega_L \\ \omega_R \end{bmatrix}$$

**A/B Coefficients:** Derived from 110 kg mass and wheel inertia.
**K Coefficient:** Represents horizontal viscous friction.

# Control Systems




**Real-time encoder measurements of the left and right drive wheels.**
**The measured wheel speeds remain close to the target speed throughout the experiment, demonstrating stable closed-loop motor control.**

# Control Systems

```
PERIYOD(150ms) -> Sol Tık: 6 | Sag Tık: 6 | Hız Farkı (Sol-Sag): 0 | Aktif PWM -> Sol: 60 Sag: 98
PERIYOD(150ms) -> Sol Tık: 2 | Sag Tık: 5 | Hız Farkı (Sol-Sag): -3 | Aktif PWM -> Sol: 60 Sag: 86
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 8 | Hız Farkı (Sol-Sag): -4 | Aktif PWM -> Sol: 60 Sag: 74
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 4 | Hız Farkı (Sol-Sag): 0 | Aktif PWM -> Sol: 60 Sag: 62
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 4 | Hız Farkı (Sol-Sag): 0 | Aktif PWM -> Sol: 60 Sag: 60
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 4 | Hız Farkı (Sol-Sag): 0 | Aktif PWM -> Sol: 60 Sag: 60
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 6 | Hız Farkı (Sol-Sag): -2 | Aktif PWM -> Sol: 60 Sag: 60
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 6 | Hız Farkı (Sol-Sag): -2 | Aktif PWM -> Sol: 60 Sag: 60
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 4 | Hız Farkı (Sol-Sag): 0 | Aktif PWM -> Sol: 60 Sag: 60
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 4 | Hız Farkı (Sol-Sag): 0 | Aktif PWM -> Sol: 60 Sag: 60
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 4 | Hız Farkı (Sol-Sag): 0 | Aktif PWM -> Sol: 60 Sag: 60
PERIYOD(150ms) -> Sol Tık: 4 | Sag Tık: 4 | Hız Farkı (Sol-Sag): 0 | Aktif PWM -> Sol: 60 Sag: 60
```

**The serial monitor displays the left and right encoder tick counts, calculated speed difference, and the PWM values applied to each motor during forward motion.**

Atalay Sunar, Çağlayan Eşmekli, Hayal Sarıkoca
The serial monitor displays the left and right encoder tick counts, calculated speed difference, and the PWM values applied to each motor during forward motion.

# Control Strategy (Proportional Closed-Loop Control)

**Control Equation**

$$PWM = PWM + K_p \times (Target - Measured)$$

```
solPWM += KP * (target - solSpeed);
sagPWM += KP * (target - sagSpeed);
```

✔️ P control was sufficient to maintain the target speed (2 km/h).

✔️ Encoder resolution is 16 ticks/revolution, making derivative control sensitive to noise.

✔️ The cumulative PWM update provides integral-like behavior.

✔️ Simpler implementation with stable real-world performance.

# Code Architecture / Firmware Logic

## Encoder-Based Speed Control and MPU6050 Turn Control

Serial Command
f / r / l
↓
Command Debounce
300 ms
↓
Motion State
Forward / Right / Left / Stop
↓
Encoder + MPU6050 Feedback
↓
PWM Output to BTS7960
↓
Left & Right DC Motors

```
HEDEF_HIZ_KMH = 2.0;
TEKER_CAP_M = 0.56;
ENC_TIK_DEVIR = 16;

hedefTikSn() =
(v / wheel_circumference) * encoder_ticks
```

```
solPWM += KP * (hedef - solHizF);
sagPWM += KP * (hedef - sagHizF);
```

```
Turn control:
MPU6050 yaw feedback
DONUS_ACI = 70°
DONUS_PWM = 80
```

**Target Speed Calculation**

$$v = 2.0 \text{ km/h} = \frac{2.0}{3.6} = 0.556 \text{ m/s}$$

$$C = \pi D = \pi \times 0.56 = 1.759 \text{ m}$$

$$\text{rev/s} = \frac{v}{C} = \frac{0.556}{1.759} = 0.316 \text{ rev/s}$$

$$\text{ticks/s} = 0.316 \times 16 = 5.06 \text{ ticks/s}$$

**Target encoder speed:**

$5.06 \text{ ticks/s}$

The serial monitor displays the left and right encoder tick counts, calculated speed difference, and the PWM values applied to each motor during forward motion.

# Karşılaşılan Zorluklar ve Çözümler

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| Sorun | Çözüm |
| --- | --- |
| Sinyal 0'a kırpılıyordu | Pasif bias devresi: baseline ~1.65 V'a çekildi |
| Sağ ↔ sol ayrışmıyordu | Dönüş-şekli özellikleri (sol'da keskin dip) eklendi |
| Dönüş açısı tutarsızdı | Encoder yerine MPU6050 jiroskop (gerçek açı) |
| Sağ teker hızlı, eğri gidiyordu | Her teker bağımsız kapalı-çevrim hız kontrolü |

# Summary

**Achievements**

- Independent mobility for disabled users
- EOG + AI human-machine interface
- Multi-disciplinary engineering experience



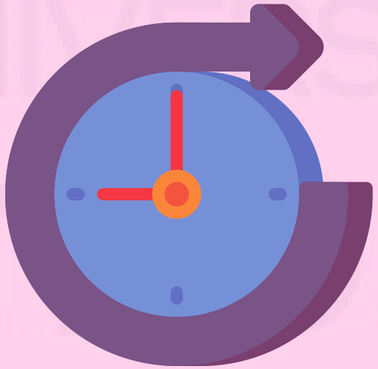
**Outcomes**

- Eye direction detection accuracy: 95.9%
- Reaction time < 1s
- Stable movement & safe obstacle stop
- Comfortable and intuitive user control



**Future Work**

•İki kanallı EOG → sağ/sol'u polariteyle daha sağlam ayırma
•Yüksek çözünürlüklü encoder → daha hassas hız
•Gömülü sistem (PC'siz) çalışma
•Çok kullanıcılı veri seti ile genelleme testi



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# THANK YOU

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